

Swift / ...

Protocol

CustomStringConvertible

A type with a customized textual representation.

iOS 8.0+ | iPadOS 8.0+ | Mac Catalyst 13.0+ | macOS 10.10+ | tvOS 9.0+ | visionOS 1.0+ | watchOS 2.0+

protocol CustomStringConvertible

Overview

Types that conform to the CustomStringConvertible protocol can provide their own representation to be used when converting an instance to a string. The String(describing:) initializer is the preferred way to convert an instance of any type to a string. If the passed instance conforms to CustomStringConvertible, the String(describing:) initializer and the print(_:) function use the instance's custom description property.

Accessing a type's description property directly or using CustomStringConvertible as a generic constraint is discouraged.

Conforming to the CustomStringConvertible Protocol

Add CustomStringConvertible conformance to your custom types by defining a description property.

For example, this custom Point struct uses the default representation supplied by the standard library:

```
struct Point {
    let x: Int, y: Int
}

let p = Point(x: 21, y: 30)
print(p)
// Prints "Point(x: 21, y: 30)"
```

After implementing the description property and declaring CustomStringConvertible conformance, the Point type provides its own custom representation.

```
extension Point: CustomStringConvertible {
    var description: String {
        return "\(x), \(y)"
    }
}

print(p)
```